RF-Based Drone Identification Using Machine Learning: Presence, Family and Operating-Condition Classification

Utkarsh Sinha
Dept. of Electronics and Telecommunication
BMS College of Engineering
Bengaluru, India
1BM23ET055

Vishal Joshy
Dept. of Electronics and Telecommunication
BMS College of Engineering
Bengaluru, India
1BM23ET058

Yash Gupta
Dept. of Electronics and Telecommunication
BMS College of Engineering
Bengaluru, India
1BM23ET061

Dr. C Gururaj
Professor
Ph.D., M.Tech
Dept. of Electronics and Communication
BMS College of Engineering
gururaj.tce@bmsce.ac.in

***Abstract*—Detecting and identifying drones from the radio-frequency (RF) signals they exchange with their controllers is a passive alternative to vision- or radar-based surveillance that works without line of sight or favourable lighting. This project builds a machine-learning pipeline that uses the public DroneRF dataset to answer three questions from a recorded RF segment: is a drone present, which drone family is it, and which recorded operating condition is most likely. Raw captures from the 2.4 GHz ISM band are segmented and converted into a 2048-point power-spectrum feature vector using the Fast Fourier Transform across the low and high RF sub-bands. Four classical classifiers — Logistic Regression, Random Forest, Support Vector Machine with an RBF kernel, and XGBoost — are compared on each task under a leakage-aware evaluation that groups all segments of one recording into the same fold. On a held-out evaluation, drone *presence* is detected with a macro-F1 of up to 1.000, drone *family* (background, Parrot Bebop, Parrot AR, DJI Phantom) reaches a macro-F1 of 1.000 for XGBoost, and the harder 10-class *operating-condition* task reaches a macro-F1 of 0.492 for XGBoost under 5-fold cross-validation, clearly above the $\sim$0.37 linear and bagging baselines. A noise-robustness test shows family accuracy degrades from 0.900 on clean data to 0.294 at -5 dB SNR, quantifying sensitivity to the Wi-Fi and Bluetooth interference that shares the band. The system is intended as an interpretable, low-cost, dataset-based decision-support tool and not as a deployed counter-drone system.**

***Index Terms*—Drone detection, RF fingerprinting, UAV identification, machine learning, DroneRF, XGBoost, feature extraction, power spectral density, data leakage.**

I. Introduction

Unmanned aerial vehicles (UAVs), or drones, are now inexpensive and widely available, and their misuse raises privacy, safety and security concerns around airports, public events and restricted areas. Reliable detection and identification of an intruding drone is therefore an important problem.

Visual and acoustic sensing require line of sight and favourable conditions, and active radar can be costly. Drones, however, continuously communicate with their flight controllers over wireless RF links, usually in the 2.4 GHz industrial, scientific and medical (ISM) band. Passively listening to these RF signals allows detection without cooperation from the drone and without line of sight.

Knowing only that *some* drone is present is of limited value. A useful system should also indicate *which* drone and, ideally, *what* it is doing, since different models and flight modes imply different threats. This project therefore develops a machine-learning pipeline that uses recorded DroneRF signals to perform three progressively harder classification tasks — presence, family and operating condition — and compares four classical machine-learning algorithms on each.

II. Overview

The system analyses recorded RF segments and produces three predictions. The three classification targets are defined as:

$$\text{presence} \in \{0,1\}, \quad \text{family} \in \{0,1,2,3\}, \quad \text{mode} \in \{0,\ldots,9\}.$$

where *presence* is drone vs. no-drone, *family* distinguishes background and three drone types, and *mode* identifies the specific recorded operating condition.

The basic operation of the system can be summarized as: **Raw RF Captures $\rightarrow$ Segmentation $\rightarrow$ FFT Feature Extraction $\rightarrow$ Label Decoding $\rightarrow$ Model Training (4 classifiers) $\rightarrow$ Leakage-Aware Evaluation**

The data comes from a single public source:

- The DroneRF dataset [1], [2].

Each capture is labelled by a 5-digit *BUI* code embedded in its filename. All three targets are decoded from this single code, so no manual labelling is required. Because each target has a small number of discrete classes, every task is a classification problem, not a regression problem.

III. Problem Statement

The presence of a drone cannot be reliably read from a single sensor in all conditions, yet knowledge of it is essential for security and airspace management. RF sensing is passive and works without line of sight, but real-world signals in the

2.4 GHz band are shared with Wi-Fi and Bluetooth, which makes the classification task noisy.

In addition, RF classifiers trained and tested carelessly can report optimistic results because many segments extracted from one physical recording are highly correlated; if such segments are split across training and test sets, the model effectively sees the test data during training. Therefore, there is a need for a data-driven system that classifies drone presence, family and operating condition from easily computed spectral features, and whose evaluation explicitly avoids this data leakage.

The proposed system uses classical machine learning to classify RF segments from the DroneRF dataset, and evaluates the trained models under a recording-grouped split and under added noise.

IV. Literature Survey

Many studies have investigated RF-based drone detection and identification using machine learning. Early work established the dataset and deep-learning baseline; later work explored feature engineering, interference robustness and lightweight deployment.

A. Dataset and Deep-Learning Baseline

Al-Sa'd et al. [1] introduced the DroneRF dataset and trained deep neural networks for detection, type and mode classification, establishing the three-task structure used here. The accompanying data descriptor [2] documents the file structure and BUI labelling scheme.

B. Classical Machine Learning Approaches

Al-Emadi et al. [3] applied XGBoost to the same three tasks and reported average accuracies of 99.96%, 90.73% and 70.09% for detection, type and mode respectively, confirming that tree-boosting is strong on this data. Ezuma et al. [4] classified UAV controllers from RF fingerprints using statistical features and classifiers such as k-nearest neighbours.

C. Interference and Robustness

Ezuma et al. [5] studied detection and classification in the presence of Wi-Fi and Bluetooth interference. Recent journal work continues this theme: robustness of RF UAV detectors [7], ensemble fingerprinting under jamming [8], and low-SNR recognition with complex-valued networks [9].

D. Feature Extraction Alternatives

Medaiyese et al. [6] used wavelet-transform features, and the LUMEN framework [10] used power-spectral-density features similar to those used here. These works motivate the spectral feature choice of this project.

E. Comparative Analysis

Table I summarizes representative works against the proposed system.

TABLE I
Literature Comparison

Reference	Method	Year	Focus
Al-Sa'd et al. [1]	Deep neural network	2019	DroneRF dataset + 3 tasks
Al-Emadi et al. [3]	XGBoost	2020	Detect / type / mode
Ezuma et al. [5]	ML + interference	2020	Robustness to Wi-Fi/BT
Medaiyese et al. [6]	Wavelet + ML	2022	Feature extraction
LUMEN [10]	PSD + lightweight net	2026	Edge deployment
Proposed	**4 classical ML, FFT features**	**2026**	**Leakage-aware 3-task comparison**

F. Research Gap

Much existing work either focuses on deep learning or reports high accuracy without explicitly controlling for segment-level data leakage. This project addresses the gap by comparing four interpretable classical models across all three tasks under a recording-grouped split, and by quantifying noise robustness directly.

V. Libraries and Tools Used

Python 3.11. Python is the core language used for preprocessing, feature extraction, model training, evaluation and plotting.

- **NumPy** — FFT, array operations and the power-spectrum feature computation.
- **Pandas** — loading and organizing the feature table, and writing it to a columnar Parquet file.
- **scikit-learn** — Logistic Regression, Random Forest, SVM, the grouped train/test split (`GroupKFold` / `GroupShuffleSplit`) and the metrics (precision, recall, F1, confusion matrix).
- **XGBoost** — the gradient-boosted tree classifier.
- **Matplotlib** — confusion-matrix heatmaps and the noise-degradation curve.
- **Kaggle API** — reproducible download of the DroneRF dataset.

VI. System Design and Methodology

Fig. 1 shows the proposed pipeline.

A. Raw Data

The DroneRF dataset contains recorded RF activity for background (no drone) and three drones: Parrot Bebop, Parrot AR and DJI Phantom. Each capture is stored as two CSV files, a low sub-band (`L`) and a high sub-band (`H`), of the 2.4 GHz band. There are 227 recording pairs in total.

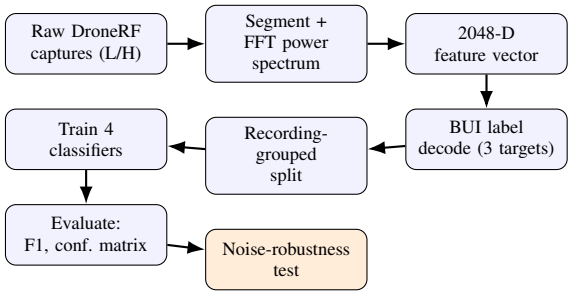


Fig. 1. System flow diagram.

TABLE II
BUI CODE TO LABELS

BUI code(s)	Family	Meaning
00000	0	Background, no drone
10000–10011	1	Bebop (4 modes)
10100–10111	2	AR drone (4 modes)
11000	3	Phantom (1 mode)

B. Label Decoding

Each file name begins with a 5-digit BUI code. The code encodes all three targets simultaneously, as summarized in Table II. Presence is 0 for background and 1 otherwise; family maps the code to one of four classes; mode is a distinct index for each of the ten BUI codes.

C. Feature Extraction

Each capture is divided into non-overlapping segments of $L = 100{,}000$ samples. For a segment x, the mean is removed and a 2048-point FFT magnitude is computed, keeping the upper half ($M/2 = 1024$ bins):

$$X_f = \left| \mathrm{FFT}_M(x - \bar{x}) \right|, \qquad f = \tfrac{M}{2} + 1, \ldots, M.$$

The low- and high-band spectra are concatenated with a continuity scaling and squared to give a power spectrum, producing a 2048-dimensional feature vector per segment. The complete dataset yields 22,700 feature vectors.

D. Leakage-Aware Split

Because all segments of one capture are highly correlated, the evaluation groups every segment of a recording into the same fold using `GroupKFold` (5 folds) or a single 75/25 grouped split. This prevents segments from one recording appearing in both training and test sets.

E. Classifiers

Four models are trained and compared on each task: Logistic Regression (a linear baseline), Random Forest (a bagging ensemble of trees), SVM with an RBF kernel (for curved class boundaries, trained on a 5000-sample subset for tractability), and XGBoost (gradient-boosted trees).

TABLE III
MACRO-F1 BY MODEL AND TASK

Task	LR	RF	SVM	XGBoost
Presence	1.000	0.978	1.000	1.000
Family	0.999	0.898	0.977	**1.000**
Mode (5-fold)	0.375	0.378	0.160*	**0.492**

*SVM trained on a 5000-sample subset per fold.

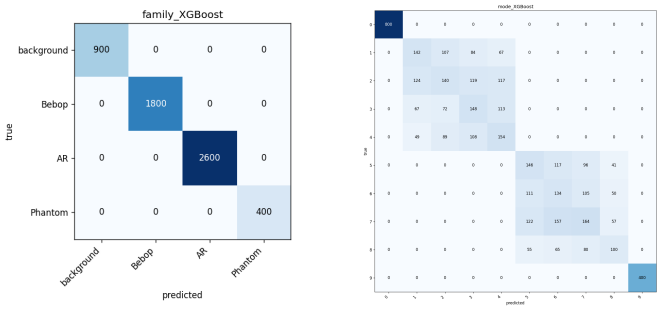


Fig. 2. XGBoost confusion matrices: family (left, perfect) and mode (right, 10-class).

F. Evaluation Metrics

For each model and task we report macro-averaged precision, recall and F1, and a confusion matrix. Macro-F1 is used as the headline metric because the classes are imbalanced:

$$\mathrm{F1} = \frac{2\,PR}{P+R}, \qquad \text{macro-F1} = \frac{1}{K}\sum_{k=1}^{K} \mathrm{F1}_k.$$

VII. RESULTS AND DISCUSSION

Table III reports macro-F1 for every model and task. Presence and family are reported on a single grouped split; the harder mode task is reported as mean $\pm$ standard deviation over 5 folds.

Presence detection is essentially solved by all models. Family classification is also solved by XGBoost, whose confusion matrix (Fig. 2) is a perfect diagonal. The 10-class mode task is substantially harder; XGBoost reaches a macro-F1 of 0.492, clearly above the linear and bagging baselines around 0.37, making it the best model on the task that matters most.

A. Noise Robustness

To test the interference concern of the 2.4 GHz band, Gaussian noise was added to the test features at decreasing SNR for the family task (Table IV, Fig. 3). Accuracy falls from 0.900 on clean data to about 0.50 at 20 dB and 0.29 at -5 dB, showing that spectral features alone are not robust to heavy noise and that explicit interference handling would be needed for deployment.

B. Fair-Evaluation Note

The near-perfect presence and family scores are honest given the recording-grouped split, but residual correlation between captures of the same mode may remain. The small 5-fold standard deviation (± 0.009) indicates the results are stable rather than the product of a single lucky split.

TABLE IV
FAMILY MACRO-F1 VS. SNR (RANDOM FOREST)

SNR	clean	20 dB	10 dB	5 dB	0 dB	-5 dB
Macro-F1	0.900	0.496	0.424	0.376	0.320	0.294

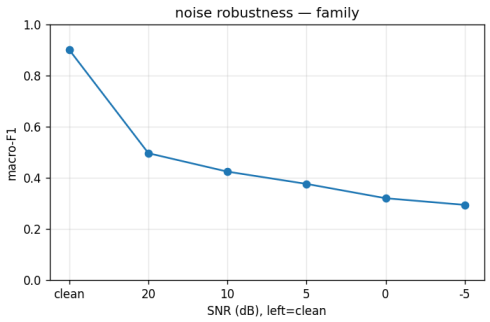


Fig. 3. Noise-robustness curve for the family task.

VIII. CONCLUSION

This project built and evaluated a classical machine-learning pipeline for RF-based drone identification on the DroneRF dataset. Four models were compared across presence, family and operating-condition classification under a leakage-aware split. Presence and family are reliably classified, while the 10-class operating-condition task remains hard, with XGBoost giving the best macro-F1 of 0.492. A noise-robustness test quantified the sensitivity of the spectral features to interference. The work shows that interpretable, low-cost classical models remain competitive on RF drone classification, and identifies operating-condition classification and noise robustness as the main directions for future work.

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